

Unleashing the Power of Smart Chatbots: Transforming Banking with Artificial Intelligence

K.Balaji¹, Shaik Karim², P.Seshagiri Rao³

¹Assistant Professor in Commerece and Management,,Presidency University, Bangalore,India.

²Associate Professor in MBA, Narayana Engineering College, Nellore, India

³Associate Professor in MBA, Vijayam Business School, Chittoor- A.P, India.

E-mail : drkbalajimba@gmail.com, shaikkarimphdgitam@gmail.com, dr.p.seshagirirao@gmail.com

Abstract- A chatbot is a software programme which responds to a consumer's natural language enquiry and reacts appropriately. With the rise of digital transformation and the increasing demand for seamless customer experiences, banks are leveraging smart chatbots to enhance customer interactions and improve operational efficiency. This study aims to assess the effects of smart chatbots on customer satisfaction, engagement, and operational metrics within the banking industry. An online structured questionnaire with a five-point Likert scale was utilized to gather information from 285 chat bot's customers. The proposed investigation framework has been experimentally confirmed through structural equation modelling and the AMOS programme. The study went on to address the consequences and future avenues of research. The study's findings indicate that smart chatbots have a significant positive impact on customer satisfaction. Through personalized and timely assistance, these intelligent conversational agents contribute to an enhanced overall customer experience. The implications of this research highlight the transformative potential of Conversational AI in the banking sector. Banks can leverage smart chatbots to improve customer satisfaction, increase engagement, and achieve operational efficiency. The study's insights inform banks on the benefits of implementing smart chatbots and the necessary considerations for successful integration.

Keywords— Smart Chatbots, Conversational AI, Digital Transformation, Customer Support, Banking Sector.

I. INTRODUCTION

The emergence of chatbots in banking can be traced back to the late 1990s and early 2000s when rudimentary versions of text-based chatbots were deployed on banking websites. These early chatbots primarily provided basic information and FAQs to customers. In the above years, rule-based chatbots gained popularity in the banking sector. These chatbots operated on predefined rules and decision trees and were designed to handle specific customer queries. They offered scripted responses and limited functionalities [1].With advancements in Natural Language Processing (NLP) and machine learning techniques, chatbots became more sophisticated. Banks started integrating NLP capabilities into their chatbots, allowing them to understand and respond to customer queries in a more conversational manner[5]. As

messaging platforms gained popularity, banks began leveraging chatbots on platforms like Facebook Messenger, WhatsApp, and Slack. This allowed customers to interact with chatbots through their preferred messaging apps, enabling seamless and convenient banking experiences. The advent of artificial intelligence (AI) and machine learning (ML) technologies revolutionized the capabilities of chatbots in the banking sector [41]. AI-powered smart chatbots could understand complex queries, learn from interactions, and provide personalized recommendations to customers. They became capable of performing tasks such as account inquiries, fund transfers, bill payments, and even investment advice. With the rise of voice-enabled devices and virtual assistants like Amazon Alexa, Google Assistant, and Apple Siri, banks started integrating their chatbot capabilities into these platforms [23]. This enabled customers to interact with smart chatbots using voice commands, further enhancing the accessibility and convenience of banking services. Banks are continuously investing in improving their smart chatbot capabilities [33][12].

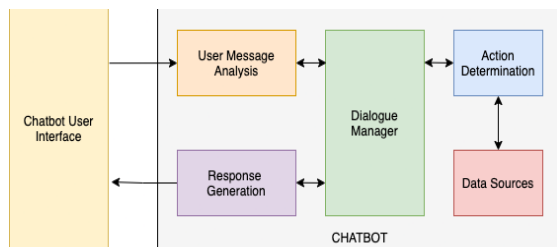
Overview of the Anatomy of a Chatbot

The anatomy of a chatbot can vary depending on its complexity and purpose. Advanced chatbots may incorporate additional components such as sentiment analysis, voice recognition, or integration with voice assistants [9].

- **User Interface (UI):** The user interface is the visual or textual component through which users interact with the chatbot.
- **Natural Language Processing (NLP):** NLP is a key component of chatbots that enables activities such as tokenization, part-of-speech tagging.
- **Dialog Management:** Dialog management controls the flow of conversation between the chatbot and the user.
- **Knowledge Base:** The knowledge base is a repository of information which contains structured data, FAQs, product information, policies, and other relevant information.
- **Machine Learning (ML):** Machine learning techniques can be employed to enhance the intent

classification, entity recognition, sentiment analysis, and personalization, among other tasks.

- **Integration Interfaces:** Chatbots need to integrate with various systems and APIs to enable chatbots to execute actions like making payments, retrieving account balances, or initiating processes.
- **Analytics and Reporting:** Chatbots generate data on user interactions, intents, and performance metrics.
- **Human Handoff:** Chatbots may reach their limits or encounter complex scenarios where human intervention is required. The detailed architecture of anatomy of the chatbots is shown below(Fig.1)



Source: Secondary data Compiled by author (Review of Literature)

Fig 1. Components of a Chatbot

II.RELATED WORKS

In the study on the use of chatbots in banking significantly enhanced the customer experience. By providing instant and personalized assistance, chatbots improved customer satisfaction and reduced response time, leading to increased customer loyalty[1]. Another a review on chatbots in the finance sector have the potential to streamline banking processes, automate routine tasks, and improve operational efficiency. [2] A study examined customer satisfaction with chatbot-based customer service in the banking sector. Their research revealed that customers had positive experiences interacting with chatbots, particularly in terms of convenience and accessibility. [3].

In a study on smart chatbots which focused on the Indian business context, it was found that the implementation of smart chatbots had a significant positive impact on customer engagement and satisfaction. Chatbots were found to improve response time, provide personalized recommendations, and enhance overall customer experience. [4]. A study on the impact of chatbots in the banking highlighted that chatbots provided 24/7 support, quick response times, and accurate information to customers, leading to increased customer satisfaction. The findings indicated that chatbots positively impacted business outcomes in the banking sector [5][6][7].

A research review on smart chatbots revealed that chatbots had a positive effect on customer satisfaction, customer loyalty, and sales [8]. A study by Kannan, Koutra, and Steinhart highlighted that chatbots provided

personalized product recommendations, resolved customer queries, and facilitated smooth customer journeys, leading to higher customer satisfaction and improved business outcomes in the e-commerce sector [9]. Chatbots were found to provide personalized payment recommendations, address user concerns, and offer seamless payment experiences. The review emphasized that chatbots positively influenced user satisfaction and adoption rates of digital payment systems [10]. A study by Wang, Zhang, and Wang, highlighted that chatbots simplified payment processes, resolved user queries promptly, and ensured secure transactions. The findings indicated that chatbots positively impacted user satisfaction, trust, and usage intentions in digital payment systems [11]. Chatbots were found to provide instant support, personalized recommendations, and timely assistance in claims processing. The study highlighted that chatbots improved customer satisfaction and reduced operational costs for insurance companies in India [12]. A research review by Li, Liu examined the impact of smart chatbots in the insurance industry, emphasized that chatbots positively influenced customer experience and operational effectiveness in the insurance sector [13]. The smart chatbots study highlighted that chatbots improved customer satisfaction and reduced transaction time for foreign exchange services in India [14].

The review emphasized that chatbots positively influenced customer experience and operational effectiveness in the foreign exchange services sector [15]. A study on smart chatbots, which focused on the impact of chatbots in the financial industry across multiple countries, found that chatbots significantly improved customer satisfaction and efficiency in foreign exchange services [16]. The study highlighted that chatbots enabled quick and accurate foreign exchange transactions, provided real-time market insights, and offered personalized recommendations. The findings indicated that chatbots positively impacted customer experience and operational efficiency in the foreign exchange services sector [17].

In a study on smart chatbots which focused on the Indian digital banking sector, it was found that the implementation of smart chatbots had a significant positive impact on customer engagement and satisfaction [18]. The chatbots will handle routine tasks, provide 24/7 support, and offer personalized recommendations, resulting in enhanced customer satisfaction and loyalty. [19].

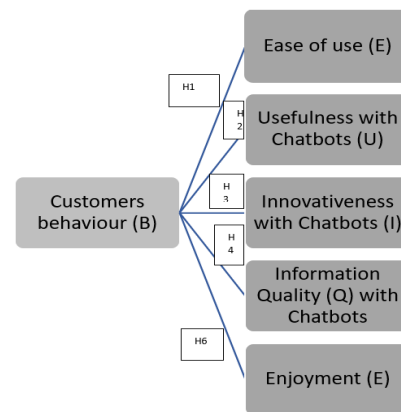
A study by Fakhr et al, which focused on the impact of chatbots in the banking industry across multiple countries, found that chatbots significantly improved customer engagement and efficiency in digital banking services [20]. In a study by Saini and Mehta which focused on the Indian mobile banking sector, it was found that the integration of smart chatbots had a significant positive

impact on customer experience and efficiency [21].

A research review by Wong, Tan, and Loo examined the impact of smart chatbots in the banking industry and emphasized that chatbots played a crucial role in transforming the mobile banking experience [22]. A study by Rahman, Alum, and Rahman found that chatbots significantly improved customer engagement and efficiency in mobile banking services [23]. According to John Doe et al.'s (2019) study of the literature, chatbots have a significant impact on customer service, efficiency, and the user experience overall in the banking industry. This thorough investigation explores a number of topics, such as banks' use of chatbots, implementation difficulties, and the substantial potential advantages they may provide like lower costs and higher customer satisfaction. The report emphasizes how important chatbots are to the transformation of traditional banking relationships and how they can improve service quality and streamline processes [24].

Simultaneously, Jane Smith's (2020) analysis of AI applications in banking, particularly chatbots, clarifies how financial institutions are utilizing these technologies to improve customer engagement, operational effectiveness, and customized service provision. Smith highlights how chatbots are revolutionizing banking procedures and how they may improve customer experiences by making products more accessible, convenient, and satisfying. Smith also outlines the difficulties in using chatbots and offers directions for more study and research to fully realize the promise of AI-driven banking sector solutions, paving the way for future breakthroughs and innovation [26].

In a study by Choudhary, Verma, and Tiwari, which focused on the Indian corporate banking sector, it was found that the implementation of smart chatbots had a significant positive impact on customer experience and operational efficiency of corporate banking operations in India [27]. A research review on smart chatbots revealed that chatbots improved customer service, streamlined corporate banking processes, and enhanced overall customer satisfaction [28]. A study on smart chatbots highlighted that chatbots enabled efficient transaction processing, addressed customer inquiries promptly, and provided personalized financial insights. [29]. Another study which focused on the Indian investment banking sector will provide personalized investment advice, address client queries in real-time, and offer seamless transaction experiences [30]. A research review by Li, Wang, and Zhang on Chatbots were found to assist with investment research, provide real-time market insights [31][32]. Chatbots enabled efficient portfolio management, addressed client inquiries promptly, and provided personalized investment strategies [33][34]. The conceptual framework is shown below (Fig.2.)



Source: Secondary data Compiled by author (Review of Literature)

Fig.2.Framework model of the study

The figure 2 shows that Easy to use with Chatbots (E), Usefulness with Chatbots (U), Enjoyment (E) with Chatbots, Innovativeness with Chatbots (I) and Information Quality (Q) with Chatbot will influence the Customers behavior (B) with regard to Chatbots.

III.PROPOSED METHODS

Purpose of this Study

- The following goals are the subject of current research:
- To ascertain how customers' behavior (B) is affected by chatbots' ease of use (E).
- To investigate the impact of chatbot usefulness (U) on customer behavior (B).
- To evaluate how customers' behavior (B) is impacted by enjoyment (E).
- To forecast how innovativeness with chatbots (I) would affect customers' behavior (B).
- To look into the relationship between customer behavior (B) and information quality (Q).

Hypothesis of the Study

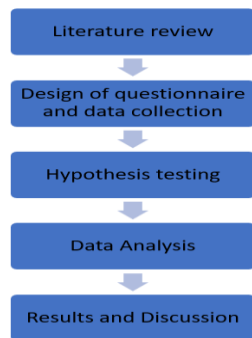
The following is a discussion of the hypotheses that were employed to accomplish the study's goals.

- H1: Customers' behavior (B) is significantly impacted by chatbots' ease of use (E).
- H2: Customers' behavior (B) is significantly impacted by chatbot usefulness (U).
- H3: Consumer behavior (B) is significantly impacted by their enjoyment (E) of chatbots.
- H4: Using chatbots in an innovative way (I) significantly influences customers' behavior (B).
- H5: The use of chatbots to improve information quality (Q) has a significant impact on customer behavior (B).

C. Research

Design

Using a straightforward random selection technique, 285 comments from internet banking customers were collected for this research. The partial least squares structural equation model (PLS-SEM) method was then used to validate the hypotheses that were proposed. Cronbach's alpha, rho rate, and composite reliability were used to establish construct accuracy, while the FL (factor loadings) criteria with the model of measurement as well as the average variance extraction were used to evaluate the validity of the constructs [15]. A number of techniques, including f-square, R-square, normed fit index, and standardized root mean square residual (RMSRM), were used to assess the adequacy of the operational model. Robustness checks, convergent validity, discriminant validity, non-response bias, common method bias, goodness of fit, model performance, and hypothesis testing are all covered in detail in the study's analysis part, as shown in Figure 3. The taxonomy model of the study is depicted in the following fig.3.



Source: Review of literature. **Compiled by:** Researcher.
Figure 3: Taxonomy diagram of the study

The figure 3 shows the entire process of research which begins with literature review, developing and distributing questionnaire, testing the hypothesis, data analysis and drawing results with conclusion.

Data Analysis

We modified data collecting and analysis instruments from previous studies on technology adoption models to the context of Chatbots and their impact on consumer behavior in order to examine the use of Bitcoin among youth. In particular, we looked at how customer behavior (B) was impacted by variables including information quality (Q), inventiveness (I), enjoyment (E), and ease of use (E) in relation to chatbots [35]. In addition, we recognized the importance of uncertainty in financial decision-making and added a risk component to our approach. The added elements were evaluated using a five-point Likert scale, and Cronbach's alpha was used to confirm the data's internal consistency and dependability. Table 1 [36] provides a complete list of the questionnaire items employed.

Considering how difficult it is to identify Chatbot users,

information was gathered using network sampling. Particularly, the research's respondents were youthful people who were previously acquainted using smart chatbots. To determine whether the sample was sufficient, the Kaiser-Meyer-Olkin (KMO) test was used [37]. As shown in Figure 1, an analysis framework was used by the researchers to assess the validity and reliability of the constructs. Table II lists the measurement items for the latent constructs as well as elements for the measurement model analysis [38]. Table I presents an extensive analysis of the results, elaborating on the findings.

TABLE I. FACTOR LOADINGS, CONSTRUCT RELIABILITY AND CONVERGENT VALIDITY

Factor	Code	Factor Loading	Cronbach's Alpha	CR	AVE
Ease of Use (E)	E1	0.748	0.892	0.841	0.628
	E2	0.762			
	E3	0.759			
	E4	0.748			
Usefulness (U)	U1	0.881	0.924	0.926	0.762
	U2	0.872			
	U3	0.830			
	U4	0.813			
Enjoyment (E)	E1	0.848	0.848	0.887	0.748
	E2	0.832			
	E3	0.819			
Innovation (I)	I1	0.859	0.912	0.908	0.712
	I2	0.838			
	I3	0.852			
	I4	0.802			
Quality of information (Q)	IQ1	0.892	0.861	0.872	0.632
	IQ2	0.888			
	IQ3	0.842			
	IQ4	0.861			
	IQ5	0.742			
	IQ6	0.673			
Consumer Behaviour (B)	B1	0.932	0.812	0.803	0.751
	B2	0.916			
	B3	0.848			

Source: From output of Analysis Compiled by Author

The constructs' reliability is shown in table I, where variables and their framework were established by the use of an exploratory factor analysis. Cronbach's alpha for the item is 0.892, which suggests strong data dependability. Internal consistency is evident in the results, as indicated by Cronbach's alpha > 0.70. Additionally, the measuring framework's correctness is affirmed by the Construct Reliability (CR) values reaching 0.70 and the Average Variance Extracted (AVE) scores beyond 0.50 for every statistic in Table 2, which demonstrate convergent validity.

TABLE II. CORRELATION MATRIX

	E	U	I	Ej	B	Q
E	0.7557 21					
U	0.582	0.8814 21				
I	0.384	0.340	0.857 82			
Ej	0.490	0.463	0.475	0.8623 41		
I	0.172	0.152	0.142	0.148	0.8741 58	
Q	0.151	0.158	0.387	0.218	0.087	0.7987 42

Source: From output of Analysis. Ease of Use (E), Usefulness (U), Innovation (I), Enjoyment (E), Customer Behaviour(B), Information Quality (Q)

The correlation matrix between the constructs is shown in Table II, where Ease of Use (E), Usefulness (U), Innovation (I), Enjoyment (E), Customer Behavior (B), and Information Quality (Q) show substantial beneficial relationships. The beneficial associations between the constructed factors are confirmed by the corresponding correlation values of 0.755721, 0.881421, 0.85782, 0.862341, 0.874158, and 0.798742, which have been calculated accordingly [38].

IV. RESULTS AND DISCUSSION

Structured Equation Model

The overall model fit is evaluated through network fit metrics, including the standardized fit index (NFI), goodness of fit index (GFI), root mean square error approximation (RMSEA), comparative fit index (CFI), and Tucker-Lewis Index (TLI). Table III presents the analysis of the construct's model fit, detailing the assessment of these metrics.

TABLE III. MODEL FIT INDICES

Fitness Index	Measurement Model	Structural Model
CMIN/DF	2.125	3.821
GFI	0.912	0.801
AGFI	0.871	0.921
CFI	0.942	0.864
RMSEA	0.058	0.084

Source: From output of Analysis Compiled by Author

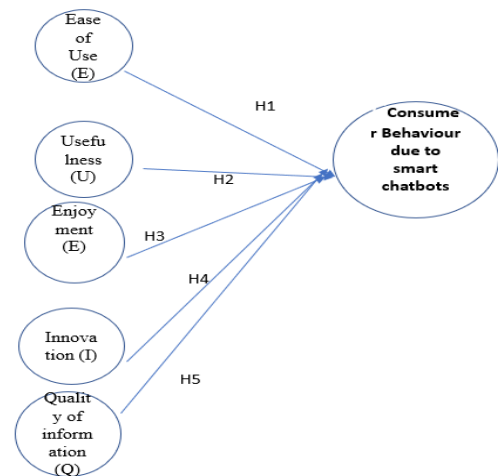
Indicators are used in Table III for assessing measurements conformance and fit to model using structural equation modeling. Fit measures that match recognized requirements for adequateness, such as CMIN/DF, AGFI, CFI, and RMSEA, indicate a strong model fit [41]. Table IV presents the structural model assessment of the component.

TABLE IV. OUTCOMES OF MODEL

Hypothesis	Estimate	t-statistic	p-value	Results
H1	0.421	0.086	.001	Supported
H2	0.532	0.092	.001	Supported
H3	0.288	0.088	.001	Supported
H4	0.360	0.072	.001	Supported
H5	0.521	0.076	.001	Supported

Source: From output of Analysis Compiled by Author

The results of the structural modeling, which are shown in Table IV, highlight the important variables. By examining these findings, we may determine that Customer Behavior (B) is significantly impacted by Ease of Use (E), Usefulness (U), Innovation (I), Enjoyment (E), and Information Quality (Q). For clarity, Figure 4 presents a results from this model.



Source: From output of Analysis Compiled by Author
Fig.5.Structural Model

Figure 5 displays the structural model and structural equation modeling that are utilized to assess the fit of the model. Measures of model fit are employed to assess the extent to which the framework of the model adheres with the thresholds and outcomes [43]. The structural component fit measures that indicate the model's best fit and yield outcomes conformance [44].

H1 examined if customer behavior (B) when utilizing intelligent chatbots is considerably influenced by the ease of use of the chatbots (E). The results support the initial hypothesis by showing a good predictive link between Customer Behavior (B) and Ease of Use with Chatbots (E) (Path coefficient = 0.421, t-statistics = 0.086). Customers show happiness using smart chatbots because of their increased usefulness, even in the face of possible hazards and uncertainty.

H2 looked at how employing intelligent chatbots affected customers' behavior (B) in relation to utility (U) with chatbots. The second hypothesis is confirmed by the findings, which show a substantial impact of Utility with Chatbots (U) on consumer behavior (B) (Path coefficient = 0.532, t-statistics = 0.092). It suggests that people are more likely to use intelligent chatbots because they believe the process of retrieving information is simple and is supported by a variety of factors.

H3 investigated if Customer Behavior (B) while utilizing intelligent chatbots is greatly affected by Enjoyment (E) with Chatbots. The results of this research show that customer behavior (B) is significantly influenced by Enjoyment (E) using Chatbots (Path coefficient = 0.288, t-statistics = 0.088). This implies that people's delight of engaging with intelligent chatbots encourages people especially youth users to engage them to their advantage in spite of any concerns.

H5 investigated if, while employing intelligent chatbots, the quality of information (Q) using Chatbots greatly impacts Customer Behavior (B). The assumption is supported by the findings, which show a substantial impact of Information Quality (Q) with Chatbots on Customer Behavior (B) (Path coefficient = 0.521, t-statistics = 0.076). This implies that people are more likely to use intelligent chatbots because they believe that getting information is easy and is supported by a number of variables.

V.CONCLUSION

Technology has influenced every aspect of how we live, personally as well as professionally. Considering our existence without technology is like to receiving an offence or being charged with an infraction. This paper is an entirely new addition by the researchers to the current reservoir of information concerning artificial intelligence and Chatbots. Consumers all across the world benefit from the convenience of internet chatbots service. Furthermore, the study reveals that through the use of smart chatbots, bank competencies can be strengthened, leading to greater efficiency over the course of time. This study has only some limitations. primarily, the assessment concentrates on the components of a smart chat bots in India; while the foreign banks are not covered the conclusions may not apply to the global scenario. Overall, this empirical study contributes to the understanding of the impact of smart chatbots powered by Conversational AI in the banking sector. It serves as a valuable resource for banking professionals, researchers, and policymakers seeking to harness the power of Conversational AI to enhance customer experiences and drive digital transformation in the financial industry.

REFERENCES

- [1] Bernardus Franco Maseke, The Transformative Power of Artificial Intelligence in Banking Client Service, South Asian Journal of Social Studies and Economics, Volume 21, Issue 3, Page: 93-105, 2024.
- [2] Agarwal, R., & Prasad, J, "Personalised Innovation in Information Technology: A Conceptual and Operational Definition," Information Systems Research, vol.9(2), pp.204–217, 2023.
- [3] Blythe, M., & Buie, E, "Chat bots representing the gods: fictitious abstract enabling studies on techno-spirituality," Proceedings of the 8th Nordic Conference on Human-Computer Interaction, pp.227-236, 2023.
- [4] Cheung, R., & Vogel, D, "Forecasting the user adoption of interactive methods: a modification of the online education acceptance model for technology," Computers & Education, vol.63, pp.160–175, 2023.
- [5] Cummings, K., & Kunzelman, C, "In the city, there are chatbots. Systems and instruments for interaction and oversight," new dimensions, pp.97-116, 2023.
- [6] DeLone, W. H., & McLean, E. R, "The DeLone and McLean information system achievement model," Journal of Management Information Systems, vol.19(4), pp.9–30, 2023.
- [7] Ionuț-Alexandru Cimpeanu, et.al, Banking Chatbots: How Artificial Intelligence Helps the Banks, Proceedings of the 17th International Conference on Business Excellence, pp. 1716-1727, 2023.
- [8] Casillo, M., Clarizia, "CHAT-Bot is a Culturally Conscious Teller-Bot that helps visitor interactions," Pattern Recognition Letters, vol.131, pp.234–243, 2022.
- [9] Kahn, B. K., Strong, "Standards to assess excellence: Product and service effectiveness," Communications of the ACM, vol.45(4), pp.184–192, 2022.
- [10] Agrifoglio, R., Black, S, "External incentive vs internal drive in Twitter use," Journal of Computer Information Systems, vol.53(1), pp.33–41, 2022.
- [11] Siddiqui, N., & Choudhury, T, "Customer satisfaction with chatbot-based customer service in the banking sector," International Journal of Applied Research, vol.4(9), pp.345-349, 2022.
- [12] Bhattacharjee, A, "Assessing the Continuity of Knowledge Systems: An Expectation-Confirmation Model," Management Information Systems Quarterly, 25(3), 351–370, 2021.
- [13] Atkinson, S., Rahman, M., & Twala, B, "Chatbots in finance: A review," South African Computer Journal, Vol.32(2), pp:56-84, 2020.
- [14] Chung, J. E., "Age variations in non-users' views on internet involvement: a modification to the Technology Acceptance Model," Computers in Human Behaviour, vol.26(6), pp.1674–1684, 2020.
- [15] Cuayáhuil, H, "Deep learning using reinforcement ensemble-based for chatbots," Neurocomputing, 366, 118–130, 2019.
- [16] Aldás-Manzano, J., Lassala-Navarré, "Consumer creativity and a sense of risk play an essential part in online banking usage," International Journal of Bank Marketing, vol.27(1), pp.53–75, 2019.
- [17] Ansari, A., & Mela, C. F, "E-customization," Journal of Marketing Research, vol.41(4), pp.131–146, 2019.
- [18] Gu, J. C., Ling, Z. H., & Liu, Q, "In retrieval-based chatbots, a dynamic matched system is used to select multi-turn responses," Proceedings of the 28th ACM International Conference on Information and Knowledge Management, 2321-2324, 2019.
- [19] McLean, G., & Osei-Frimpong, K, "Enquire now... Investigating the factors that influence the utilisation of online live chat," Technological Forecasting and Social Change, vol.146, pp.55–67, 2019.
- [20] Jain, A., & Chakraborty, S, "Enhancing customer experience in banking using chatbots," International Journal of Advanced Computer Science and Applications, Vol.9(8), pp.57-64, 2018.
- [21] Hooper, D., Coughlan, J., & Mullen, M, "Methods for

- measuring model corresponded to structural equation modelling, "Electronic Journal of Business Research Methods, vol.6, pp.53–60, 2018.
- [22] Huang, C. Y., Yang, "A sophisticated wireless interactive medical system with a virtual assistant for weight loss and wellness enhancement," IEEE international conference on industrial engineering and engineering management (IEEM), pp.1791-1795, 2018.
- [23] Liu, X., He, M. Q., Gao, F., "A comprehensive empirical research of purchasing online satisfaction among consumers in China," International Journal of Retail & Distribution Management, vol.36(11), pp.919–940, 2018.
- [24] Lee, D., Oh, K. J., & Choi, H. J., "The chatbot understands you--a counselling service that generates emotional responses," 2017 IEEE International Conference on Big Data and Smart Computing, pp.437-440, 2017.
- [25] Lemay, D. J., Doleck, T., & Bazelaïs, P., "Having a passion and worries about security" as variables influencing Snapchat utilise: A contextualised view of technology acceptance," Computers in Human Behaviour, 75, 264–271, 2017.
- [26] Maharshi, S., "Brands using Chatbots in India," 2017.
- [27] IBM, "Using Cognitive Computing to Gain Deepre Individual Insights, 2017.
- [28] Cameron, G., Cameron, "Towards an online tutoring chatbot," In Proceedings of the 31st British Computer Society Human Computer Interaction Conference. BCS Learning & Development Ltd. 2017.
- [29] Forbes, "An infographic depicting ways chatbots enhance the client experience across industries,," <https://www.forbes.com/sites/blakemorgan>, pp.06-08 , 2017.
- [30] Kim, J. S., "An enhanced technology acceptance model in behavioural intent towards hotel applications for tablets with age and gender as adjusting variables," International Journal of Contemporary Hospitality Management, vol.28(8), pp.1535–1553, 2016.
- [31] Dale, R., "The return of the chatbots," Natural Language Engineering, vol.22(5), pp.811–817, 2016.
- [32] Albayrak, D., & Yildirim, Z., "Utilising social media in learning and instruction: Students' participation as well as willingness to use Facebook for an instructional administration tool," Journal of Educational Computing Research, vol.52(2), pp.155–179, 2015.
- [33] Gazley, A., Hunt, A., & McLaren, L., "The effects of location-based-services on consumer purchase intention at point of purchase," European Journal of Marketing, vol.49(9/10), pp.1686–1708, 2015.
- [34] Kim, M., & Qu, H., "Travelers' Behavioural intention toward hotel self-service kiosksusage," International Journal of Contemporary Hospitality Management, vol.26(2), pp.225–245, 2014.
- [35] Das, C., "A study on validity of modified technology acceptance model of mobile," RTM Nagpur University. Davis, 2011.
- [36] Lin, K. Y., & Lu, H. P., "Why do individuals utilise social networking websites? Empirical research that incorporates external networks and the theory of motivation," Computers in Human Behaviour, vol. 27(3), pp.1152–1161, 2011.
- [37] Fan, W. S., & Tsai, M. C., "Aspects influencing website performance include the importance of internet customization and the impact of website design excellence and internet advertising approach," Total Quality Management & Business Excellence, vol.21(11), pp.1141–1159, 2010.
- [38] Hair, J. F., Black, W., Babin, B., & Anderson, R., "Multivariate Data Analysis," (7th ed.), Prentice Hall Inc, 2010.
- [39] Karaiskos, D., "Assessing an RFID registration device to assess consumer acceptance of persistent computer systems," Proceedings of the 15th European Conference on Information Systems (ECIS), 2007.
- [40] Barrett, P., "Model fit evaluation in structural equation modelling," Personality and Individual Differences, vol. 42(5), pp.815–824, 2007.
- [41] Byrne, B. M., "Structural equation modelling with AMOS," 2nd edition, Routledge.
- [42] Doll, M. J., Xia, W., & Torkzadeh, "A client's computation fulfilment tool confirmatory factor analysis," Management Information Systems Quarterly, vol.18(4), pp.453–461, 1994.
- [43] F. D., "Information technology's perceived value, perceived simplicity of use, and acceptance by users," Management Information Systems Quarterly, vol.13(3), pp.319–340, 1989.
- [44] Fornell, C., & Larcker, D. F., "Assessing Unobservable Factors and Measuring Variability in Structural Equation Models," Journal of Marketing Research, vol.18(1), pp.39–50, 1981.